Agentic Privacy Control Center

Sebastian Ferragut sferragut@ucsd.edu
Nian-Nian Wang niw002@ucsd.edu
Jesse Huang jeh027@ucsd.edu
Jimmy Huang jih103@ucsd.edu
Arya Verma a5verma@ucsd.edu
Mentor: Haojian Jin haojian@ucsd.edu
Mentor: Qiyu Li qil016@ucsd.edu

UC San Diego | SCHOOL OF COMPUTING, INFORMATION AND DATA SCIENCES | HALICIOĞLU DATA SCIENCE INSTITUTE

1. Problem: Privacy Settings Are Hard to Navigate

Digital platforms often bury privacy and data-sharing controls behind nested menus, inconsistent labels, and confirmation flows. This creates friction for users trying to understand or adjust their account's:

- tracking/personalization and ads,
- third-party sharing,
- audience/visibility,
- account deletion and retention controls.

Objective: An end-to-end system that automatically discovers, structures, visualizes, and safely modifies privacy settings across heterogeneous websites.

2. Automated Web Crawler + Screenshot Collection

Workflow: Navigates the settings pages and captures all user-configurable controls through screenshots and click depths.

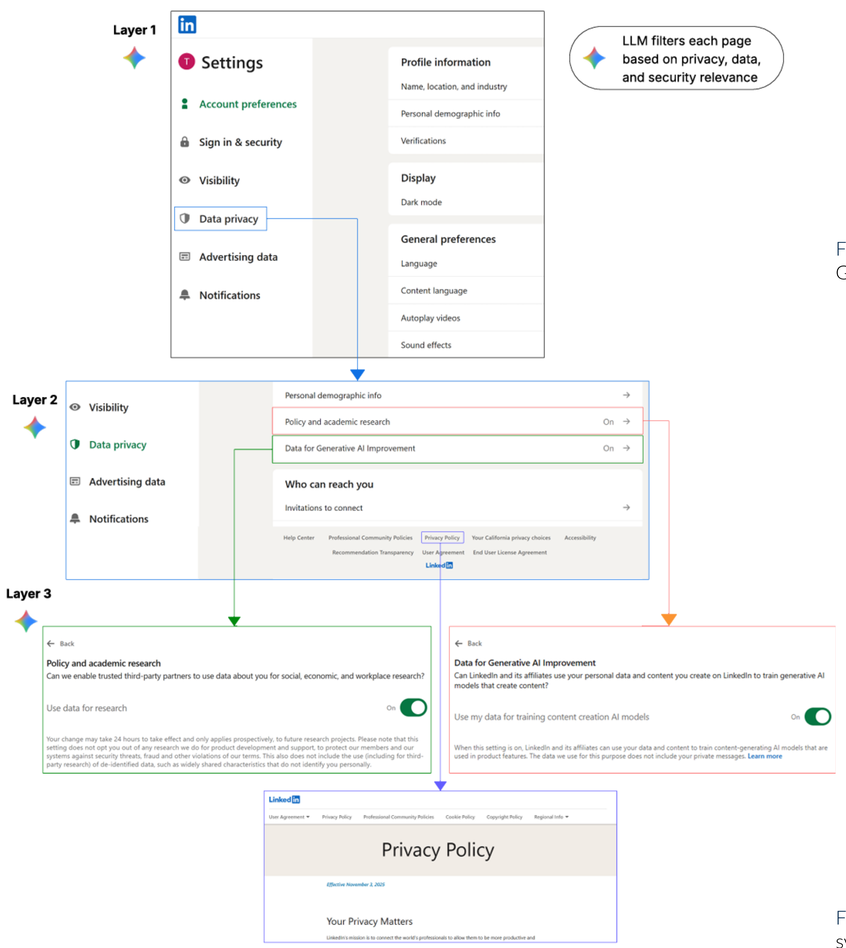

Figure 1. Demonstrates **web crawler flow** as it navigates LinkedIn's settings page using a breadth-first search approach, stopping when user-configurable control page is reached. Each **layer** of the graph reflects the **number of clicks** a user must make to reach a specific page.

3. Dataset: Extraction + Normalization + Classification

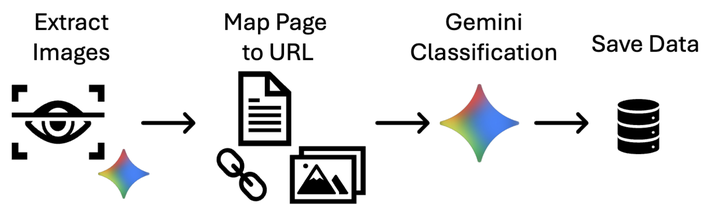

Figure 2. Pipeline that extracts settings information from crawler screenshots, **classifies each setting into semantic categories,** and outputs per-setting records in a CSV for downstream analysis, visualization, and actuation downstream.

4. Privacy Chatbot: Planner–Executor–Verifier Actuation

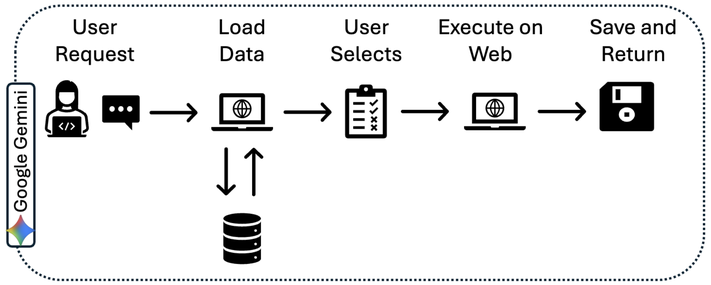

Figure 3. Architecture used to **safely modify privacy settings.** UI selection of settings allows Gemini-enabled bot to leverage database for settings changes.

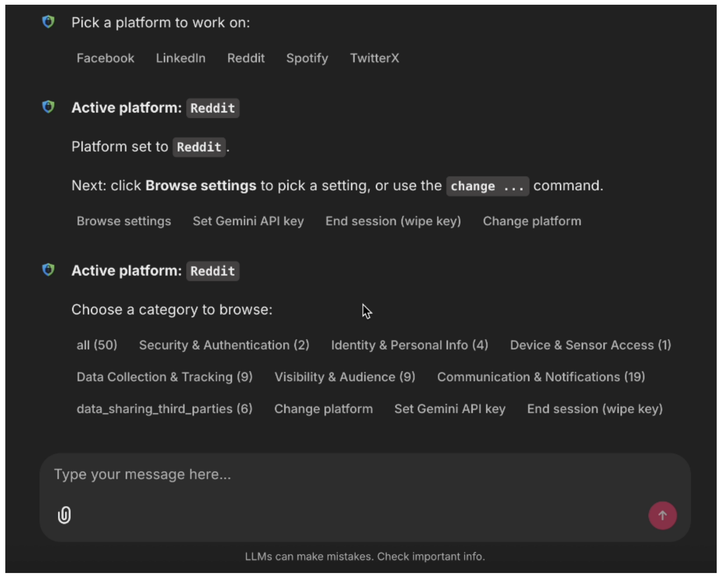

Figure 4. Agentic privacy assistant interface used to locate and modify privacy settings. The system **interprets user commands**, generates structured action plans, and executes UI interactions through a browser automation framework.

5. Takeaways From The Privacy Problem

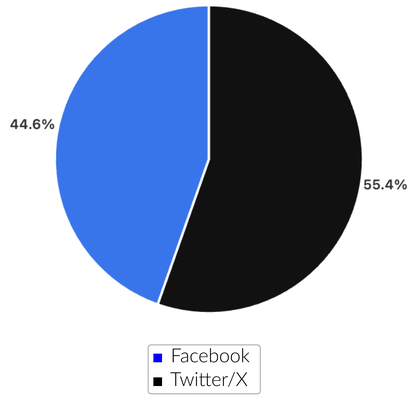

Figure 5. Click-depth comparison across platforms. Case study of **navigation effort** between Facebook and Twitter/X: the former has more settings, while the latter is more accessible.

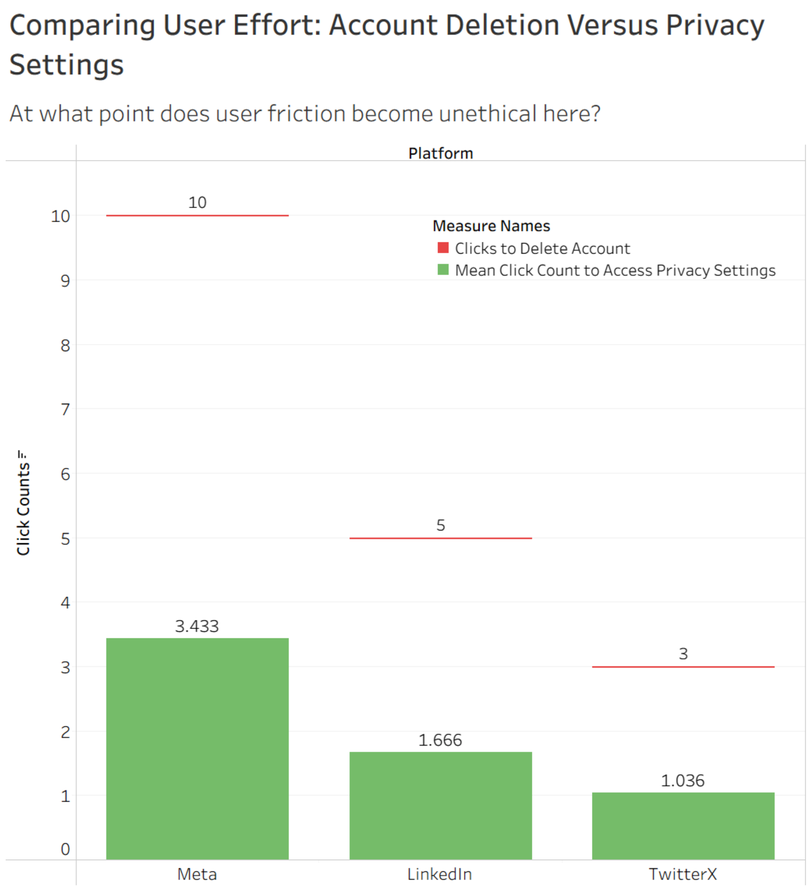

Figure 6. Comparison of user effort required to locate privacy settings across platforms. Differences in navigation depth illustrate how interface structure can **significantly increase** the steps required for users to access and modify privacy controls.